Course on Knowledge Graphs in the era of Large Language Models

Session 3: Reasoning with RDFS Semantics and OWL 2 RL

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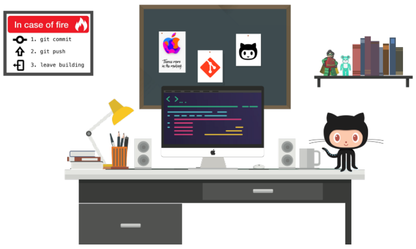
Contents

1 Git Repositories and dependencies

Support codes for the laboratory sessions are available in *GitHub*. There are two repositories, one in Python and another in Java:

`https://github.com/city-knowledge-graphs`



For Java developers we use maven to deal with dependencies (see pom file). For Python developers there is always a `requirements.txt` within each lab folder with possibly new dependencies. It is recommended to use environments, but it is not strictly necessary.

2 Inference rules (cheatsheet)

As seen in the lecture, Figure 1 summarizes the necessary inference rules we need for the lab.

Rule	If	Then add
rdf1	(x p y)	(p rdf:type rdf:Property)
rdfs2	(p rdfs:domain c) (x p y)	(x rdf:type c)
rdfs3	(p rdfs:range c) (x p y)	(y rdf:type c)
rdfs4a	(x p y)	(x rdf:type rdfs:Resource)
rdfs4b	(x p y)	(y rdf:type rdfs:Resource)
rdfs5	(p rdfs:subPropertyOf q) (q rdfs:subPropertyOf r)	(p rdfs:subPropertyOf r)
rdfs6	(p rdf:type rdf:Property)	(P rdfs:subPropertyOf p)
rdfs7	(p rdfs:subPropertyOf q) (x p y)	(x q y)
rdfs8	(c rdf:type rdfs:Class)	(c rdfs:subClassOf rdfs:Resource)
rdfs9	(c rdfs:subClassOf d) (x rdf:type c)	(x rdf:type d)
rdfs10	(c rdf:type rdfs:Class)	(c rdfs:subClassOf c)
rdfs11	(c rdfs:subClassOf d) (d rdfs:subClassOf e)	(c rdfs:subClassOf e)
rdfs12	(p rdf:type rdfs:ContainerMembershipProperty)	(p rdfs:subPropertyOf rdfs:Member)
rdfs13	(x rdf:type rdfs:Datatype)	(x rdfs:subClassOf rdfs:Literal)
	(p owl:inverseOf q)	(q owl:inverseOf p)
	(p owl:inverseOf q) (x p y)	(y q x)
	(p rdf:type owl:SymmetricProperty) (x p y)	(y p x)

schema only
data + schema
not relevant

Figure 1: Figure adapted from "Towards Efficient Schema-Enhanced Pattern Matching over RDF Data Streams". International Semantic Web Conference (ISWC) 2011. Slides.

3 RDFS inference

Consider the following set of triples (we will refer to them as the graph $\mathcal{G}_{\text{rdfs}}$).

```
@PREFIX : <http://city.ac.uk/kg/lab4/>
@PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
@PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
:Person     a                    rdfs:Class .
:Man        a                    rdfs:Class ;
            rdfs:subClassOf      :Person .
:Woman      a                    rdfs:Class ;
            rdfs:subClassOf      :Person .
:Parent     a                    rdfs:Class ;
            rdfs:subClassOf      :Person .
:Father     a                    rdfs:Class ;
            rdfs:subClassOf      :Parent ;
            rdfs:subClassOf      :Man .
:Mother     a                    rdfs:Class;
            rdfs:subClassOf      :Parent ;
            rdfs:subClassOf      :Woman .
:Child      a                    rdfs:Class ;
            rdfs:subClassOf      :Person .
:hasParent  a                    rdf:Property ;
            rdfs:domain          :Person ;
            rdfs:range           :Parent .
:hasFather  a                    rdf:Property ;
            rdfs:subPropertyOf   :hasParent ;
            rdfs:range           :Father .
:hasMother  a                    rdf:Property ;
            rdfs:subPropertyOf   :hasParent ;
            rdfs:range           :Mother .
:isChildOf  a                    rdf:Property ;
            rdfs:domain          :Child ;
            rdfs:range           :Parent .
:Ann        a                    :Person ;
            :hasFather           :Carl ;
            :hasMother           :Juliet .
```

3.1 Manual inference

Task 3.1 Decide if $\mathcal{G}_{\text{rdfs}}$ derives the following statement(s) and explain *why* or *why not*. In the positive case, then indicate the RDFS inference rules from the lecture (also found at `http://www.w3.org/TR/rdf-mt/`) to prove your answer. If the statement is not derived, then explain, informally or formally, why this is so. Formally can be done via a counterexample, *i.e.*, with an interpretation that entails $\mathcal{G}_{\text{rdfs}}$, but it does not the statement.

Statement 1 `:Father rdfs:subClassOf :Person .`

Statement 2 `:Woman rdfs:subClassOf :Person .`

Statement 3 `:Juliet a :Person .`

Statement 4 `:Ann a :Child .`

Statement 5 `:Ann :isChildOf :Carl .`

Statement 6 `:Ann :hasParent :Juliet .`

Statement 7 `rdfs:range rdf:type rdfs:Resource .`

Statement 8 `:Mother rdfs:subClassOf :Person .`

3.1.1 Example solution

Example solution for Statement 1:

```
:Father rdfs:subClassOf :Person .
```

True, the statements is derived by $\mathcal{G}_{\text{rdfs}}$. `:Father` is (transitively) a subclass of `:Person`. Rule **rdfs11**. Statements 1 and 2 below are found in $\mathcal{G}_{\text{rdfs}}$ and are premises to the application of the inference rule **rdfs11**, which yields the statement we're after (Statement 3).

Proof:

1. `:Father rdfs:subClassOf :Parent` — P
2. `:Parent rdfs:subClassOf :Person` — P
3. `:Father rdfs:subClassOf :Person` — 1, 2, rdfs11

In the proof above each line is marked with "P" if the statement is a premise, *i.e.*, exits in $\mathcal{G}$, or with the rdfs rule and the line identification of the input statements.

3.2 RDFS inference programmatically

We are using the OWL-RL python library which builds on top of RDFLib and has an RDFS reasoning component.[1] The file in GitHub `RDFSReasoning.py` expands our example graph $\mathcal{G}$ using the RDFS inference rules. A Jupyter notebook is also provided.

Task 3.2: Check if the statements in **Task 3.1** are True or False over the extended graph or model (*i.e.*, the graph after applying reasoning). The graph $\mathcal{G}_{\text{rdfs}}$ is provided within the file `lab-rdfs.ttl`.

4 OWL 2 RL entailment

Consider the following set of triples (we will refer to them as the graph $\mathcal{G}_{\text{owl2rl}}$).

[1] Documentation: `https://github.com/RDFLib/OWL-RL`

```
@PREFIX : <http://city.ac.uk/kg/lab4/>
@PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
@PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@PREFIX owl: <http://www.w3.org/2002/07/owl#> .
:Person     a                   owl:Class .
:Man        a                   owl:Class ;
            rdfs:subClassOf     :Person .
:Woman      a                   owl:Class ;
            rdfs:subClassOf     :Person .
:Parent     a                   owl:Class ;
            rdfs:subClassOf     :Person .
:Father     a                   owl:Class ;
            rdfs:subClassOf     :Parent ;
            rdfs:subClassOf     :Man .
:Mother     a                   owl:Class;
            rdfs:subClassOf     :Parent ;
            rdfs:subClassOf     :Woman .
:hasChild   a                   owl:ObjectProperty ;
            owl:inverseOf       :hasParent .
:hasParent  a                   owl:ObjectProperty ;
            rdfs:domain         :Person ;
            rdfs:range          :Parent .
:hasFather  a                   owl:ObjectProperty ;
            rdfs:subPropertyOf :hasParent ;
            rdfs:range          :Father .
:hasMother  a                   owl:ObjectProperty ;
            rdfs:subPropertyOf :hasParent ;
            rdfs:range          :Mother .
:Ann        a                   :Person ;
            :hasFather          :Carl ;
            :hasMother          :Juliet .
```

4.1 Manual entailment

Task 3.3. Indicate if the following statements are derived by $\mathcal{G}_{\text{owl2rl}}$. $\mathcal{G}_{\text{owl2rl}}$ is within the OWL 2 RL profile so one could apply, among many others, similar inference rules to those for RDFS.[2] Indicate in your proof which are the involved triples from $\mathcal{G}_{\text{owl2rl}}$.

Statement 1 `:Juliet :hasChild :Ann .`

Statement 2 `:Ann a :Child .`

4.2 OWL2 RL entailment programmatically

We are using the OWL-RL python library. The file `OWLReasoning.py` in GitHub expands our example graph $\mathcal{G}_{\text{owl2rl}}$ using OWL 2 RL reasoning. A Jupyter notebook is also provided.

[2] `https://www.w3.org/TR/owl2-profiles/#Reasoning_in_OWL_2_RL_and_RDF_Graphs_using_Rules`

Task 3.4. Check programmatically if the above statements (Task 3.3) are True or False over the extended graph (*i.e.*, after applying reasoning). The graph $\mathcal{G}_{\text{owl2rl}}$ is provided within the file `lab-owl2rl.ttl` in the corresponding `lab4` folders.

5 Solutions

5.1 Task 3.1

Statement 2

`:Woman rdfs:subClassOf :Person .`

True. This triple is explicitly stated in $\mathcal{G}$.

Statement 3

`:Juliet a :Person .`

True. Proof:

1. `:Ann :hasMother :Juliet .` — P
2. `:hasMother rdfs:range :Mother .` — P
3. `:Juliet rdf:type :Mother` — 1, 2, rdfs3
4. `:Mother rdfs:subclassOf :Woman` — P
5. `:Woman rdfs:subclassOf :Person` — P
6. `:Mother rdfs:subclassOf :Person` — 4, 5, rdfs11
7. `:Juliet rdf:type :Person` — 3, 6, rdfs9

Statement 4

`:Ann a :Child .`

False. It seems intuitive that `:Ann` is a child, but there is not connection between `:Ann` and `:Child` that can be entailed. To get this inference, `:hasFather` and/or `:hasMother` should be declared as sub-property of `:isChildOf`.

Statement 5

`:Ann :isChildOf :Carl .`

False. Similarly to Statement 4.

Statement 6

`:Ann :hasParent :Juliet .`

True. Proof:

1. `:Ann :hasMother :Juliet` — P
2. `:hasMother rdfs:subPropertyOf :hasParent` — P
3. `:Ann :hasParent :Juliet` — 1, 2, rdfs7

Statement 7

`rdfs:range rdf:type rdfs:Resource .`

True. This statement is an axiomatic triple and is always satisfied in RDFS models.

Statement 8

`:Mother rdfs:subClassOf :Person .`

True. Similarly to Statement 1.

5.2 Task 3.2

Solutions in GitHub. The solutions use `ASK` queries (we will learn about them in our Session 5) over the extended graph. Check the solution files in the GitHub repository.

5.3 Task 3.3

Statement 1

`:Juliet :hasChild :Ann .`

True. Proof:

1. `:Ann :hasMother :Juliet` — P
2. `:hasMother rdfs:subPropertyOf :hasParent` — P
3. `:hasChild owl:inverseOf :hasParent` — P
4. `:Ann :hasParent :Juliet` — 1, 2, rdfs7[3]
5. `:Juliet :hasChild :Ann` — 3, 4, prp-inv2[4]

Statement 2

`:Ann a :Child .`

False. Ann is the child/daughter of Carl and Juliet but there is not a class `:Child`. For this statement to hold, the range of `:hasChild` should be `:Child`.

5.4 Task 3.4

Solutions in GitHub. As for the RDFS case, the solutions use `ASK` queries over the extended graph. Check the solution files in the GitHub repository.

[3]Or *prp-spo1* with respect to OWL 2 RL reasoning.

[4]OWL 2 RL reasoning: `https://www.w3.org/TR/owl2-profiles/#Reasoning_in_OWL_2_RL_and_RDF_Graphs_using_Rules`